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Project Information Document (PID)

Appraisal Stage | Date Prepared/Updated: 20-Jan-2025 | Report No: PIDIA00880



BASIC INFORMATION

A. Basic Project Data

Project Beneficiary(ies)	Region EASTERN AND SOUTHERN AFRICA	Operation ID P505094	Operation Name Digital Zambia Acceleration Project (DZAP)
Financing Instrument Investment Project Financing (IPF)	Estimated Appraisal Date 13-Jan-2025	Estimated Approval Date 27-Mar-2025	Practice Area (Lead) Digital Development
Borrower(s) Republic of Zambia	Implementing Agency Smart Zambia Institute		

Proposed Development Objective(s)

Increase access to, and inclusive use of, the internet and digitally enabled services.

Components

1. Affordable broadband
2. Interoperable and secure data platforms
3. Digital Skills and Capabilities
4. Program management and capacity building

PROJECT FINANCING DATA (US\$, Millions)

Maximizing Finance for Development

Is this an MFD-Enabling Project (MFD-EP)? Yes

Is this project Private Capital Enabling (PCE)? Yes

SUMMARY

Total Operation Cost	120.00
Total Financing	120.00
of which IBRD/IDA	100.00
Financing Gap	0.00

DETAILS



World Bank Group Financing

International Development Association (IDA)	100.00
IDA Grant	100.00

Non-World Bank Group Financing

Commercial Financing	20.00
Unguaranteed Commercial Financing	20.00

Environmental And Social Risk Classification

Moderate

Decision

The review did authorize the team to appraise and negotiate

1. Zambia will be the fifth operation of the Inclusive Digitalization in Eastern and Southern Africa (IDEA) Multiphase Programmatic Approach (MPA) Project approved by the World Bank's Executive Directors on June 27, 2024, following Republic of Angola, Democratic Republic of Congo (DRC), Republic of Malawi, and the Common Market for Eastern and Southern Africa (COMESA) as the coordinating Regional Economic Community (REC).

B. Introduction and Context

Country Context

2. **The Republic of Zambia is a low-income country of around 20 million people which faces multiple challenges to its human, social, and economic development.** Zambia's growth has been volatile, achieving middle-income status in 2011 and then slumping by 2019 to the lowest growth in two decades, compounded further by the COVID-19 pandemic. Zambia was reclassified as low income in July 2022 for one year. Most Zambians are employed in subsistence agriculture and the informal services sector, where wages are low. The lack of diversification within and outside of mining and agriculture, both highly vulnerable to shocks, caused economic growth to slow to 3.1 percent per year between 2015 and 2019, mainly attributed to falling copper prices and declines in agricultural output due to increased rainfall variability. The latest period of drought, which began in January 2024, is considered the worst for two decades and, because the country is so dependent on hydropower, it has caused regular loadshedding (20+ hours daily) which negatively impacts digital services¹. Nevertheless, copper production output increased by 6.3 percent in the first half of 2024, primarily driven by rising prices and favorable mining conditions associated with drought.

¹ The drought has affected the Central, Copperbelt, Eastern, Lusaka, Northwestern, Southern, and Western provinces. On February 29, 2024, Zambia declared a national emergency due to the drought, which was caused by climate change and El Niño. <https://reliefweb.int/disaster/dr-2024-000018-zmb>



3. **Zambia's young and growing population presents significant opportunities and challenges for job creation and more inclusive growth.** Zambia's population is young and growing rapidly, with 46 percent under 15 years and about 20 percent between 15 and 24 years. The young population means the country is well-placed for a digital leap, but translating this into a demographic dividend will necessitate substantial investments in human capital to create a healthy and skilled labor force, particularly for women, to enable inclusive growth. Although Zambia is one of the most politically stable countries in Africa, weak institutions and limited accountability have hindered service delivery and more inclusive growth. Weak policy coordination and implementation of reforms, as well as limited transparency and accountability, have created space for vested interests to exercise a disproportionate influence in policy-making and allocation of public resources, thereby hindering pro-poor growth. On the positive side, the format of institutions in Zambia (laws, policies, systems, and structures) has shifted to mirror global best practices. The new Government has taken bold steps to stabilize the economy, refocus on private sector-led growth, restore public confidence, and strengthen institutions. Key measures have targeted the debt problem through improved fiscal management, scaling up social safety nets programs, and taking service delivery and resources closer to the people by increasing financing to constituencies while strengthening local governance and accountability.

Sectoral and Institutional Context

4. **Digitalization can be a powerful tool for development, transforming entire sectors and delivery of services as well as creating employment.** Higher levels of internet use are associated with significant developmental gains—increases in the number of jobs² and a reduction in poverty. Digital can contribute to improved productivity and entrepreneurship. But the current level of use in the Eastern and Southern Africa region is inadequate and uneven. By putting in place a foundational basis for inclusive high-speed broadband networks, secure data hosting, digitally verifiable identification (ID), digital financial services (DFS), and data sharing, countries are establishing the highways of tomorrow that are needed to build inclusive and dynamic digital economies, societies, and governments. The benefits of getting digital public infrastructure (DPI)³ right can be enormous: empowered people, improved access to services and economic opportunities for all, more efficient delivery of social assistance, improved data for decision-making and innovation unlocked⁴. Regulatory harmonization to extend trust across borders can widen the benefits of high-speed broadband, DPI, DFS and citizen-centric e-Government services to facilitate cross-border trade.

5. **Digitalization can also play a critical role in tackling challenges, such as climate change, gender inequality and health emergencies.** Zambia's population is highly exposed to many climate hazards, the most common being floods and droughts (see annex 1). Digital technologies⁵ can help build resilience to the effects of climate change, for instance, by supporting early warning systems for extreme weather events and by developing secure systems for DFS to facilitate electronic cash transfers and remittances.⁶ They also provide a platform for women to access knowledge, resources, and essential services,⁷ especially when supported by appropriate training and strategies. The use of accessible DFS, such as mobile money, has the potential to revolutionize financial services. Digital technologies can also expand access and

² World Bank. 2023. *Digital Africa: Technological Transformation for Jobs*.

³ DPI refers to a set of foundational systems and their organizational frameworks (for example, laws and institutions) that enable core functions in today's digital age, including as core elements the ability to verify identities, send and receive money, and exchange data.

⁴ See World Bank. 2024. *Zambia Public Finance Review*, Report No. 193530, which highlighted significant data discrepancies from different sources, and governance issues. There is a need to enhance digital administrative data for policy-making.

⁵ Digital technologies include mobile money, emergency communication facilities, internet access, including from schools, government offices, and free wireless fidelity (WiFi) zones, and so on.

⁶ When there are floods and certain areas are inaccessible by road, mobile money platforms, such as Zambia's popular service providers like MTN Mobile Money and Airtel Money, facilitate rapid and efficient delivery of financial assistance during climate-related disasters. This digital financial infrastructure enables emergency funds to reach affected communities swiftly, supporting recovery efforts and reducing the economic impact of disasters. Source: [Digital Frontiers Institute](#)

⁷ The project aims to improve the internet experience for 3.7 million Zambian women through new and enhanced use.



improve efficiencies and effectiveness of emergency response and aid delivery in fragile contexts as well as to vulnerable communities in remote, rural areas.

6. **Although Zambia has made significant strides on its path to digital transformation, there remain key challenges to inclusivity that the project will address.** These include gaps in internet coverage particularly in rural areas where cell towers have not yet been upgraded to support mobile broadband and fixed line networks are largely absent. Cross-border infrastructure between countries is missing, which is hindering the development of seamless digital trade. There are notable disparities in internet access by women, persons with disabilities (PwD), and older people. Gaps in energy access are a further constraint and efforts to expand internet coverage will be aligned with expansion of energy access.⁸ On the services side, the lack of digitization of government records⁹ and public services means that citizens often have to travel considerable distances and queue to access basic services. Digital skills are also lacking, meaning that the country is not well placed to take advantage of new technologies, such as artificial intelligence (AI) and machine learning. The project will tackle three critical challenges of connectivity, digital services and digital skills, which correspond to the three main components of the project.

7. **Although Zambia has a relatively competitive telecommunications market, connectivity is lagging the regional average (Component 1).** With four licensed mobile operators (Airtel, MTN, Zamtel, and Zedmobile), at least six internet service providers (ISPs), and an experienced regulatory authority, Zambia Information and Communications Technology Authority (ZICTA), the country benefits from a favorable environment for further network expansion. The information and communication technologies (ICT) sector has experienced double digital growth and has shown resilience despite the shocks affecting the wider economy. However, the further expansion of usage is constrained by low incomes and relatively high prices, compounded by high excise duties on imported ICT equipment. Internet penetration, at around 21 users for every 100 inhabitants, is below the Eastern and Southern Africa regional average of 28 percent. Although prices for mobile services have become more affordable in recent years, owing to a competitive market, the price of fixed broadband services (5 GB per month) costs more than 12 percent of gross national income (GNI) per capita.¹⁰ However, the market entry of Starlink, a low-earth orbit (LEO) satellite provider in October 2023, is bringing greater price competition and providing new connectivity options in rural areas. The Government's broader goal is to increase internet penetration to 80 percent of the adult population, by 2026 as per the 8th National Development Plan (8NDP), though in 2021 the figure stood at only 21.2 percent, one of the lowest in the region. This will require a significant private sector investment to extend the current 10,000 km of fiber optic cable, and the project aims to add at least a further 2,000 km.

8. **Digital transformation in Zambia will strengthen the economy and create more equitable society by extending the range of digital services available (Component 2).** Digital technologies have a role to play in achieving many of the country's development outcomes. Zambia will increasingly need to rely on digital technologies to ensure that public services, businesses, and individuals are accessible to all. Additionally, these technologies will help withstand current and future hazards and develop a set of resilience measures, to ensure business continuity of government and avoid interruptions in service delivery. A combination of widespread access to broadband, digitally enabled services, and DFS can offer a powerful platform to remove barriers of distance and narrow gender gaps in digital skills as well as internet access and usage. It can also lower costs, improve efficiencies in the delivery of services, create new job opportunities, and enhance human development while also contributing to economic growth and reduced poverty.¹¹

⁸ Of relevance here is the *Mission 300 Initiative* which aims to connect an additional 300 million people in Sub-Saharan Africa by 2030 (see: <https://www.worldbank.org/en/programs/energizing-africa>) and the ASCENT program (P177099). The Digital and Energy teams are closely coordinating the projects in Zambia, notably at the level of digital mapping of demand and infrastructure.

⁹ Paper-based government records are critical to essential service delivery but are also highly vulnerable to climatic events such as floods in Zambia.

¹⁰ ITU. 2024. *ICT Price Baskets*, data for 2023, at <https://www.itu.int/en/ITU-D/Statistics/Dashboards/Pages/IPB.aspx>.

¹¹ GSMA. 2024. "Driving Digitalization of the Economy of Zambia: Leveraging Policy Reforms." The Global System for Mobile communications Association (GSMA) predicts that digitalization of trade in Zambia could add a further 0.39 percent to GDP by 2028 and digitalization of the mining sector would add 0.49 percent.



9. **A key element supported by the project is Digital ID.** Zambia is transitioning from its legacy paper-based National Registration Card (NRC) system, introduced before independence in 1964,¹² to a more modern and integrated digital identity ecosystem. The NRC system covered some 94 percent of the adult population by 2021,¹³ and subsequent advances through the Integrated National Registration Information System (INRIS) have registered 15 percent of adults using biometrics. However, the Government is planning to implement a homegrown foundational ID system, based on open standards, and building on existing investments. This new system will prioritize privacy, security, and interoperability, enabling the Government to manage its infrastructure independently and reduce reliance on external vendors.¹⁴ The Ministry of Home Affairs and Internal Security (MoHAIS) is collecting minimal demographic data (such as Name, Date of Birth, Address, Gender and biometric attributes --10 fingerprints and a photograph); only what is needed to support a unique ID. No sensitive or ethnic data is captured. ID enrollment is free of charge with the first issuance of the ID credential expected to be free. All citizens, marginalized communities, Refugees and Legal residents are eligible for enrollment. Through the ongoing ID4D technical assistance, the government is being supported in developing an operational plan, which includes expanding registration sites via post offices and implementing a public engagement strategy to accelerate registration campaigns.

10. **Digital ID is part of Zambia's broader vision to establish a robust DPI, with digital ID serving as a foundational building block for inclusive service delivery.** Complementary platforms -- such as ZamPay, the Government's eServices portal, ZamSign for e-Signatures, ZamConnect for data integration, ZamPay for interoperable digital payments and ZamConnect for data exchange -- are critical enablers of this vision, facilitating seamless access to public services, enhancing financial inclusion, and driving private sector innovation. ZamPortal, currently offers just 300 out of an estimated 1,500 services. The World Bank's interventions on Digital ID are guided by the *ID4D Principles*, implemented in close coordination with UNDP, UNHCR and other development partners in Zambia.¹⁵ The government plans to provide diverse authentication options, including biometrics, smart cards, QR codes, One-Time Passcodes (OTPs) and Paper-Based Credentials with Digital Features ensuring inclusivity and accessibility for all users. Through integration with various public and private services, this will enable seamless access to healthcare, financial services, and social welfare programs fostering trust and empowering individuals to manage their identity data securely. With technical assistance (TA) from the World Bank's Identification for Development (ID4D) Trust Fund,¹⁶ the Government is exploring cost-effective ID credential options, including digitally signed quick response (QR) codes, durable plastic cards, and polycarbonate cards. While significant progress has been made, the Government is still evaluating and will select the most suitable official ID credential to align with its comprehensive DPI strategy.

11. **Enhanced connectivity and increased access to digital services requires a focus on building digital literacy and advanced digital skills (Component 3).** The National Electronic Government Plan 2023-2026¹⁷ emphasizes the need to increase the consumption of e-services by providing both basic and advanced digital skills training through Digital Transformation Centers (DTCs). The government plans to complement basic digital skills with a mobile e-learning application hosting the complete basic literacy curriculum, ensuring continuity and constant learning for citizens. Another priority is a targeted intervention to build a skilled and employment-ready workforce capable of leveraging emerging technologies responsibly. Ongoing discussions are exploring ways to link and complement these digital skills initiatives with other sectors, such as social protection and education, while identifying potential private sector partnerships to effectively scale training efforts.

¹² ID4D Diagnostic of ID Systems in Zambia. 2024.

¹³ [https://databank.worldbank.org/source/identification-for-development-\(id4d\)-data](https://databank.worldbank.org/source/identification-for-development-(id4d)-data)

¹⁴ 2013-14 Zambia Demographic and Health Survey: <https://www.dhsprogram.com/pubs/pdf/SR219/SR219.pdf>

¹⁵ <https://www.idprinciples.org/>

¹⁶ The World Bank ID4D: <https://id4d.worldbank.org/>

¹⁷ The National Electronic Government Plan (2023-2026) [https://www.szi.gov.zm/wp-content/uploads/2023/08/Final-National e-Government Plan -2023-Final-17.08.2023.pdf](https://www.szi.gov.zm/wp-content/uploads/2023/08/Final-National-e-Government-Plan-2023-Final-17.08.2023.pdf)



C. Proposed Development Objective(s)

Development Objective(s) (From PAD)

Increase access to, and inclusive use of, the internet and digitally enabled services

Key Results

12. The Results Framework is aligned with the overall IDEA MPA and is set out in table 1.

Table 1. Results Framework for DZAP

Indicator	Baseline (Dec 2022)	End Target (Oct 2030)
People using broadband internet (additional number)	0	3,400,000
• of which female	—	1,700,000
People with digitally verifiable identification (number)	1,500,000	4,000,000
• of which female	750,000	2,000,000
People using digitally enabled services (additional number)	0	2,000,000
• of which female	—	1,000,000
Volume of international data traffic: used international bandwidth in kbit/s per capita (number)	29.8	90.0

B. Project Components

13. DZAP forms the fifth operation of the IDEA MPA, which was approved by the World Bank Board on June 27, 2024, with an overall financing envelope of US\$2.48 billion. The expected allocation of US\$100m in IDA grant is shown in table 2, together with the expected contribution of US\$20m from UCF.

Table 2. DZAP Project Components and Funds Allocation

Component	MPA Pillar/Sub-pillar	IDA (US\$, millions)	UCF (US\$, millions)
1. Affordable broadband	1	50	18
1.1: Extending the backbone and regional network	1.1	20	10
1.2: Supplying last mile connectivity	1.2	25	8
1.3: Strengthening the enabling environment	1.4	5	0
2. Interoperable and secure data platforms	2	38	2
2.1: Developing digital ID services	2.1, 2.3	23	2
2.2: Enhancing digital government and services	2.2	13	0
2.3: Fostering regional trade facilitation	2.2, 2.3	2	0
3. Digital skills and capabilities	3	5	0
3.1: Enhancing basic digital literacy	3.1	2	0
3.2: Nurturing employment-ready digital skills	3.2	3	0
4. Program management and capacity building	4	7	0
Total		100	20



Component 1. Affordable broadband (US\$50 million as IDA grant, plus an estimated US\$18 million in UCF)

14. **This component focuses on expanding essential, high-quality, resilient, and affordable broadband infrastructure and services.** Activities under this pillar aim to strengthen institutions and help drive private investment in telecom networks, which are a key foundation for using digital technologies and data productively. It will do this through investments, carried out alongside the private sector, to increase access and improve network speeds and connectivity. This will be done through a One World Bank approach, in close collaboration with IFC and MIGA. Activities are outlined in the following paragraphs.

15. **Subcomponent 1.1: Extending the backbone and regional networks.** The Government plans to extend connectivity to unserved and underserved areas to close the connectivity gap. This can be incentivized by using catalytic funding (matching investments) from project funds, that will be used to fund infrastructure owned and operated by the private sector. The project goal is to add at least 2,000 km of fiber, in the backbone and last mile, and this may be supplemented by funds from the national Universal Access and Service Fund (UASF). It is estimated that US\$20 million in project funds in backbone and regional networks could leverage an additional US\$10 million in co-investment in infrastructure from network operators, wholesale infracos including ISPs, mobile network operators, fibercos, satellite operators and other investors, through an efficient allocation of public funds that will attract new market entrants. A further ambition of the Government is to enhance cross-border connectivity by ensuring there are multiple fiber optic border crossings, and resilient loops, with each of Zambia's neighbors. Three of Zambia's neighbors are already IDEA beneficiaries (Angola, DRC, and Malawi). The Government is also interested in establishing further cross-border links through Lake Tanganyika¹⁸ and the Lobito corridor and harmonizing telecom policy and regulatory frameworks to align with regional frameworks. This subcomponent will include the following activities:

- (a) Commissioning preparation of a Commercial Transaction Manual (CTM) to analyze market dynamics, advise on procurement modalities, and guide the mechanics of the competitive procurement process for component 1, by setting out procedures for the PIU to follow, considering likely risks and mitigation measures.
- (b) Undertaking a feasibility study, coordinated with other IDEA countries, to select those backbone, cross-border and cross-lake routes, that will help extend connectivity into noncommercial and underserved areas, including preparation of environmental and social impact assessments (ESIAs). The chosen firm would also monitor infrastructure deployment, but high-level network design will be responsibility of operators bidding.
- (c) Conducting a competitive, lot-by-lot bidding process to allocate project funds for infrastructure investment (see Box).
- (d) Strengthening government capacity to operate a dedicated government network using dark fiber or managed services from commercial network operators, including support for cybersecurity.
- (e) Combined efforts between the Digital and Energy teams to generate digital maps of user demand and infrastructure in Zambia, coordinate site selection, and examine potential for commercialization of fiber held by the electricity utility.

¹⁸ See planned links with Burundi, at: <https://extensia.tech/burundi-zambia-agree-underwater-fibre-connection/>



16. **Subcomponent 1.2: Supplying last mile connectivity.** This subcomponent aims to extend internet connectivity to a minimum of 500 selected government institutions, including ministries, departments, and agencies (MDAs), including schools, and higher education institutions (HEIs), especially those in climate-vulnerable hotspots.¹⁹ If funding becomes available for a second phase, connectivity support could be extended to health facilities, livestock and fisheries camps, refugee settlements and host communities, and so on. Currently just 263 out of the targeted 15,600 national sites have been connected to the Government Wide Area Network (GWAN), leaving a huge infrastructure deficit, and the supply of international internet bandwidth is just 1.8 Gbit/s for the whole of government. Although this segment of the market is less commercially attractive, an investment of US\$25 million in project funds should nevertheless be enough to leverage an additional US\$8 million in infrastructure co-investment. This investment will be used to support the private sector in rolling out mobile broadband coverage (4G/5G cellular and/or LEO satellite) and upgrading 2G cell sites to more energy-efficient 4G/5G cellular technologies, as well as installing additional internet exchange points (IXPs) and data caches, at the local level, to improve latency.²⁰ As lack of power is a major constraint, site selection will be carefully coordinated with

Box: Competitive procurement mechanisms to leverage private sector investment

In component 1 of the project, we are proposing to use public funds, from the project, to leverage matching investments from the private sector, to achieve an expanded deployment of infrastructure in the backbone, last-mile and regional networks. These models use the “build, own, operate” principle to achieve sustainability for operations and maintenance of the network. Two particular models*, widely used in other IDA-funded projects, will be employed:

- **“Matching investments”**, whereby project funds are matched by investments from the private sector. Project funds would be allocated through a multi-round reverse auction, administered over an electronic auction platform, in which each route or site to be connected would be one lot. It is anticipated that invitations to tender to the private sector will be issued with indicative IFC and MIGA terms sheets to allow private companies to seek support for their part of the matching investments.
- **“Pre-purchase of bulk internet capacity”** to stimulate demand in the public sector, including schools and HEIs, to encourage the private sector to invest, under an “anchor tenant” model.

The aim is to close the gap between the connectivity needs of rural areas and the financial commitment that network operators are willing to make for financing what might otherwise be an uncommercial investment. The means by which these competitive procurement mechanisms would be deployed will be set out in the CTM to be prepared.

Those routes and sites that benefit from a contribution from project funds would follow World Bank procurement guidelines in the bidding process and would be subject to certain conditions. These would include: (i) the use of open access principles (to allow other operators to use the infrastructure at a reasonable rate); (ii) adherence to climate-resilient standards and use of renewable energy solutions, such as solar power, for power; (iii) the application of World Bank environmental and social standards during construction; (iv) the allocation of a certain capacity (e.g., a number of fiber strands on a fiber optic cable) for the exclusive use of the Government or ZAMREN, for the lifetime of the fiber (i.e. dark fiber). The percentage to be allocated would be proportionate to the relative levels of investment by the Government and the network operator. The use of dark fiber by the Government and ZAMREN will assist with the provision of pre-paid internet capacity under sub-component 1.2; (v) a technologically-neutral bidding process allowing for bidders to propose alternative technologies to fiber in the last mile that meet a minimum required level of functionality (e.g., fixed wireless access, 5G cellular or LEO satellites); and (vi) Service level agreements to be signed between the Government and the winning bidders.

*Note: * These models described in more detail in World Bank/CEPA. 2024. [Leveraging Private Sector Investment in Digital Communications Infrastructure in Eastern and Southern Africa](https://documents1.worldbank.org/curated/en/099135507312434566/pdf/IDU1a0df719e1a15214a9f19be4123f35220101c.pdf). Available at:*

<https://documents1.worldbank.org/curated/en/099135507312434566/pdf/IDU1a0df719e1a15214a9f19be4123f35220101c.pdf>

¹⁹ Since mid-January 2024, areas in central and southern Zambia have experienced a severe dry spell, devastating nearly half of the country's maize cultivation across 1 million ha and affecting 1 million households in eight provinces, with the most severe impacts in Southern, Central, Eastern, Northwestern, Western, and Lusaka regions.

²⁰ World Bank. 2024. *Green Digital Transformation: How to Sustainably Close the Digital Divide and Harness Digital Tools for Climate Action*. Climate Change and Development Series.



the World Bank's Accelerating Sustainable and Clean Energy Access Transformation Program (ASCENT, P180547) and the International Telecommunication Union (ITU)/United Nations Children's Fund (UNICEF) GIGA initiative,²¹ for opportunities to extend combined digital and energy access to schools, HEIs, and other selected sites, using renewable energy solutions. Support to schools and HEIs would be coordinated with the Zambian Research and Education Network (ZAMREN) and aligned with eGovernment activities to be supported under Sub-component 2.2.

17. This subcomponent will include the following activities:

- (a) Selecting sites and preparing ESIs, using the same consultancy as for Subcomponent 1.1, with active participation by the respective line ministries and in close coordination with the ASCENT and GIGA teams.
- (b) Pre-purchase of bulk internet capacity, with coordination at the regional level with other IDEA beneficiaries, under long-term supply agreements (typically 5–10 years), to support the future needs of government users, including in government-supported public free WiFi hotspots, and especially those activities targeted under different project activities (notably under Subcomponents 2.2 and 3.1).
- (c) Conducting a competitive, lot-by-lot bidding process (combined with Subcomponent 1.1c above) to allocate project funds to those network operators bidding (see Box).
- (d) Supporting ZAMREN, including covering its membership fees to join the European Union (EU) Africa Connect 4 program,²² to leverage additional funds and assistance for open educational resources.

18. **Subcomponent 1.3: Strengthening the enabling environment.** This subcomponent aims to improve the overall policy and legislative framework governing digital infrastructure, promoting market competition, affordability, and quality of service. The Government is keen to explore options for broadband market development, optimization of the UASF, and key incentives to attract investments in expansion of networks and technologies such as 5G. To minimize the environmental impact of ICT infrastructure and ensure energy efficiency, the Government aims to focus on management of e-waste to promote green digital technologies, such as green data centers and satellite connectivity, as stipulated in the National e-Government plan. The Government is currently reviewing some policies including ICT policy, Electronics Communications and Transactions Act, AI, a 'dig-once' policy, and infrastructure sharing policy, and these will be supported under the project. Additionally, the project will support secure online transactions through electronic signatures, improving efficiency in public services and digital banking.²³ Additional support will be provided to strengthen trust services by operationalizing the Data Protection Commissioner (DPC), established in 2021 and currently hosted by the Ministry of Technology and Science (MoTS), and the planned Cybersecurity Agency, also established in 2021 under the Cybersecurity and Cybercrimes Act.²⁴

19. This subcomponent will include the following activities:

- (a) Assessing the current policy and regulatory landscape and institutional arrangements in the ICT sector to inform downstream investments in the project.
- (b) Supporting implementation of policy and regulatory reforms, as per the recommendations of such assessment, including in areas such as operationalization of the UASF, and adoption of dig-once and infrastructure-sharing policies.

²¹ <https://www.unicef.org/innovation/giga>

²² <https://africaconnect4.net/>. ZAMREN currently manages regional research and education network (REN) traffic for DRC, Malawi, and Zimbabwe on behalf of EU AfricaConnect 3.

²³ Digital public infrastructure (DPI) policy note on e-signatures. <https://openknowledge.worldbank.org/entities/publication/d56f94c3-c1c8-4b17-b479-fd68f9551b1c>

²⁴ It is planned that this will be updated as two separate acts, dealing, respectively, with cybersecurity and cybercrimes. Bank support will focus on the former.



- (c) Providing TA, practical support and ICT equipment for operationalization of the DPC. This will include (i) developing an implementation strategy including a governance structure; (ii) establishing standard operating procedures and sector-specific guidelines; (iii) building capacity for data controllers and licensing of data auditors and data processors including knowledge exchange to learn from advanced data protection offices; and (iv) acquiring a digital platform to manage and monitor data protection functions.
- (d) Providing TA, practical support and ICT equipment for operationalization of the Cybersecurity Agency. This will involve: (i) establishing a centralized national Security Operations Center (SOC); (ii) strengthening cybersecurity capabilities and capacities; (iii) strengthening the Cybersecurity Incident Response Team (CIRT), established in 2012, including acquiring relevant tools and firewalls to assist with incident response, such as geolocator equipment to enhance detection of telecom related fraud, vulnerability assessment and audit assurance tools, and securing the signaling system (SS7) protocol; (iv) fostering knowledge exchange and collaboration; and (v) conducting cyber awareness campaigns and consumer education.
- (e) Developing a National Strategy for AI and regulations for emerging technologies and building capabilities in emerging technologies such as big data, blockchain, machine learning, and robotics.
- (f) Developing e-waste regulations and guidelines and supporting establishment of e-waste disposal/recycling facilities.²⁵

Component 2. Interoperable and secure data platforms (US\$38 million as an IDA grant, plus US\$2 million in UCF)

20. This component will focus on deploying interoperable DPI. Notably digital ID, to enhance efficiencies in both public and private sectors, strengthen institutions, and develop cross-cutting enablers and digital safeguards needed to promote adoption of digital technologies and digitally enabled services at the national level and across borders. Activities under this component are outlined below.

21. **Subcomponent 2.1: Developing digital ID services.** The Ministry of Home Affairs and Internal Security (MoHAIS), through its Department of National Registration, Passport, and Citizenship (DNRPC), is legally mandated to issue National Registration Cards (NRCs) and collect biometric data. As part of its modernization efforts, Zambia is transitioning from a legacy paper-based ID system to a more advanced, secure, and inclusive digital identification system. Around 7.1 million out of an estimated 10 million ID records have already been digitized, but there is a need to further modernize the entire ID ecosystem in the country and advance toward a national digital ID system. With technical support from the World Bank's ID4D team, the Government has chosen to develop a homegrown digital ID enrollment system based on open standards. This approach enhances privacy and security, reduces dependency on vendors, and allows the Government to maintain control over its infrastructure, ensuring a more flexible and cost-effective system.

22. **Zambia's transition to a fully paperless system requires an upgraded digital ID framework** capable of supporting digitally verifiable credentials and authentication processes. A modernized system with enhanced biometric capabilities will ensure robust, secure, and reliable identification for all citizens, enabling seamless access to digital services and transactions. Integrating the digital ID with other key registries, such as the social registry and farmer registry, is critical to improving the planning and delivery of government programs, streamlining identification and verification services for beneficiaries of programs such as social protection and agricultural subsidies, including e-vouchers ensuring that services reach their intended beneficiaries efficiently. This shift will not only enhance service delivery but also reduce fraud and ensure efficient, direct delivery of benefits such as cash transfers and subsidies. By simplifying know-your-customer (KYC) processes, digital ID will significantly enhance access to financial services through empowering marginalized groups, such

²⁵ The development of e-waste regulations, guidelines, and disposal facilities will primarily integrate and focus on reuse, repair, and recycle, and will be coordinated with the Zambia Environmental Management Agency (ZEMA).



as women and rural communities to open bank accounts, access credit, and utilize digital payment systems to participate more in the digital economy.

23. **The 2024 ID4D diagnostic report outlines key legal reforms needed to strengthen Zambia's Digital Identity System.** Proposed changes include updating the National Registration Act to allow additional biometric data collection, such as iris scans, and simplifying documentation requirements for marginalized groups. The report also recommends clarifying dual citizenship and streamlining naturalization processes under the Citizenship Act, revising the Birth and Death Registration Act to include non-medical facility registrations, and harmonizing the Marriage Act for equal recognition of all types of marriage. These reforms will be supported through TA to ensure effective implementation and capacity building for stakeholders involved. This will provide expertise and resources to guide the revision of legal frameworks, develop comprehensive operational strategies, and establish mechanisms for enforcing the updated laws.

24. This **digital ID subcomponent** will include the following activities:

- (a) Competitively procuring solar-powered biometric ID registration kits.
- (b) Competitively procuring a mobile based digital authentication and e-signatures solution.
- (c) Accelerating the rollout of ID registration to an additional 2.5 million people, using an enrollment system based on open standards. The project aims to increase the number of women and girls with IDs by implementing targeted awareness campaigns, supporting enrollment initiatives, simplifying application processes with agent facilitation, and ensuring well-staffed and accessible enrollment centers.
- (d) Enhancing capacity building and knowledge exchange to ensure application of best practices in digital ID.
- (e) TA to implement key legal reforms.

25. **Subcomponent 2.2: Enhancing digital government and services.** This subcomponent will support the upgrade of ZamPay from its current focus on person-to-government (P2G) payments to an integrated government-to-people (G2P) platform, forming part of the national DPI. This will strengthen the platform's scalability, allowing it to support a range of citizen-facing programs, based on user demand, including social protection programs offering a scalable and resilient payment system, especially during periods of crisis and shock. By integrating the digital ID system with social registries and payment platforms, the project will enable efficient targeting and disbursement of social protection benefits, such as cash transfers and subsidies, directly to beneficiaries. This reduces delays and leakages, ensuring timely support for vulnerable groups. Additionally, the streamlined KYC processes enabled by the digital ID will expand access to DFS by empowering marginalized groups, such as women and rural communities, to open bank accounts, access credit, and utilize digital payment systems. The improved ID framework will also simplify authentication for e-services, enabling citizens to access services such as healthcare, financial services, education more conveniently and securely. By fostering inclusivity, this subcomponent ensures that the digital ID becomes a gateway to greater social and economic participation for all citizens. ZamSign will be enhanced and upgraded to provide a secure, reliable and robust e-Signatures solution, facilitating seamless digital signing and authentication for a wide range of transactions and services. Additionally, the project will onboard key government agencies including local authorities, agricultural institutions, and the Zambia Revenue Authority (ZRA) onto ZamConnect to enable secure data sharing between government entities. The Government is interested in leveraging cloud storage services as more government services are digitized. This will require a data classification exercise to distinguish between those few services that need localized data storage and the majority of services that do not handle sensitive data and can therefore be hosted on the cloud.

26. **This subcomponent will support digitization efforts in key sectors where digital transformation is expected to deliver significant impact.** An initial assessment of the existing digital landscape is being conducted under the PPA to identify priority government services to digitize. Additional support will be provided to digitize these services and develop transaction-based portals, facilitating seamless user interaction with the ZamPortal. Priority will be given to projects within the World Bank's portfolio where digital tools serve as critical enablers, ensuring close collaboration between the



Government and the World Bank teams, particularly in social protection and DFS, and supporting local government. Other sectors, such as health, may also be targeted at a later stage if additional IDA funding becomes available. This subcomponent will include the following activities:

- (a) Operationalizing the Government Enterprise Architecture, currently under development.
- (b) Conducting a business process reengineering of ZamPortal to streamline workflows, improve the efficiency, and accelerate adoption of government e-Services.
- (c) Upgrading ZamPay into a scalable platform, enabling secure G2P payments.
- (d) Digitizing and onboarding selected citizen-centric government services onto the ZamPortal, based on user demand.
- (e) Digitizing government legacy records for digital ID, and births and death registration, and reengineering business processes to ensure improved service delivery.
- (f) Developing a government cloud strategy to inform implementation and adopting secure cloud services.²⁶
- (g) Developing a data classification and governance framework with guidelines to migrate legacy records to digital format, including migration to cloud-based services.
- (h) Onboarding key government agencies to the government data exchange platform ZamConnect to enable secure data sharing between government entities.
- (i) Providing TA to Smart Zambia Institute (SZI)²⁷ on emerging technologies and conducting digital and information systems audits to strengthen its capacity for digital transformation.

27. **Subcomponent 2.3: Fostering regional trade facilitation.** Zambia is reforming its border operations to minimize interventions, aligned with ongoing initiatives under a Coordinated Border Management approach. This subcomponent will support Zambia's efforts to promote regional integration by providing TA aimed at increasing efficiencies at the border points and moving toward 'no-stop' borders along key digital smart corridors, in close cooperation with COMESA and neighboring countries. The project will support government efforts to help reform border operations and minimize interventions, facilitate cross-border data exchange, ensure mutual recognition of IDs, harmonize e-transactions, and enhance trade facilitation through automation reforms. The expected increase in regional trade is measured through the PDO indicator on regional data flows. COMESA is benefitting from a grant under IDEA Phase 1 and will support regional coordination with the Government, the **Smart Africa Trust Alliance (SATA)**, and the Southern Africa Development Community (**SADC**) while leveraging ongoing trade initiatives supported by the World Bank projects such as **Trade Facilitation Support Program (TFSP)**,²⁸ **Zambia Agribusiness and Trade Project II (ZATP II)**,²⁹ and the Transport Corridors for Economic Resilience (**TRACER**) project,³⁰ aiming to grow regional digital trade. This subcomponent, carried out in conjunction with COMESA and other IDEA countries, will include the following activities:

- (a) Conducting a feasibility study on the move toward no-stop borders. This includes automation reforms (single window, single payments, and so on), risk management, and one-stop (or no-stop) border solutions with neighboring countries to enhance trade facilitation in accordance with the World Trade Organizations Trade Facilitation Agreement.
- (b) Developing mechanisms to facilitate cross-border data exchange. The ZRA is already exchanging customs data with Botswana, Malawi, and Zimbabwe, and is currently working with DRC, Namibia, and Tanzania to

²⁶ See World Bank. 2017. *Cloud Computing Assessment and Action Plans*.

²⁷ SZI is officially called the Electronic Government Division formed by the Electronic Government Act, 2021.

²⁸ TFSP: <https://www.worldbank.org/en/programs/trade-facilitation-support-program>

²⁹ ZATP II: <https://projects.worldbank.org/pt/projects-operations/project-detail/P179507>

³⁰ TRACER project: <https://www.worldbank.org/en/news/press-release/2024/02/20/new-world-bank-grant-to-improve-transport-and-trade-connectivity-between-afe-zambia-and-tanzania> .



develop customs system interconnectivity for electronic data exchange. ZRA is keen to enhance data sharing features and expand to other trading partners, notably South Africa, to gain advance notice of goods in transit.

- (c) Piloting cross-border mutual recognition of national IDs and ID verification mechanisms.
- (d) Harmonizing e-Transactions and cross-border regulations on data protection, privacy, and exchange with clear guidelines for cross-border data flows, processing, and storage of personal and sensitive data.

Component 3: Digital skills and capabilities (US\$5 million in IDA grant)

28. **This component focuses on building digital competencies at both foundational and advanced levels, specifically targeting underserved citizens at the base of the pyramid.** By providing basic digital literacy and developing employment-ready digital skills, this component aims to empower digitally enabled citizens, promoting the productive use of digital platforms and increasing the consumption of e-services. Additional support will be provided to develop advanced digital skills, thereby strengthening the supply of talented individuals equipped with market-aligned advanced digital skills to support the development of the Zambian ICT sector. Activities under this component are outlined in the following paragraphs.

29. **Subcomponent 3.1: Enhancing basic digital literacy.** Through SZI and ZICTA, the Government plans to deploy Digital Ambassadors within communities to provide foundational digital literacy training. SZI has already piloted basic digital skills training, leveraging content and platforms from the ITU and GSMA, in four Digital Transformation Centers (DTCs) located in schools, each equipped with internet connectivity provided via Starlink satellite technology. The expanded number of DTCs will serve as hubs, providing free internet access and offering training in digital literacy, entrepreneurship, and innovation to create job opportunities. In partnership with the Ministry of Youth, Sports, and Arts (MYSA) and Zambia Postal Services (ZAMPOST), SZI is working to expand this Program. This subcomponent will support the Government's efforts to scale up the basic digital literacy trainings, while ensuring online safety, targeting underserved populations at the base of the pyramid who are predominantly women, elderly, and PwDs. Additional digital literacy training will be provided by the Digital Ambassadors, particularly young women, targeting high-impact use cases such as using digital payment platforms and enhancing cybersecurity awareness within financial services, and for beneficiaries of social protection programs. This subcomponent will include the following activities:

- (a) Adapt a basic digital literacy curriculum and training manuals from best practice resources available while integrating local content translated into local languages and embedding cybersecurity awareness and considerations in the design of the training.
- (b) Provide basic targeted digital literacy training delivered through the DTCs in priority sectors, particularly targeting beneficiaries of social protection programs.
- (c) Expand the number of DTCs, and provide ICT equipment for them, and cover ICT equipment and operational costs for the Digital Ambassadors.
- (d) Promote productive digital usage and economic empowerment for women and girls by integrating gender- and disability-sensitive approaches (such as using assistive technologies), utilizing women trainers as Digital Ambassadors, including for sites connected under component 1.
- (e) Develop a financial sustainability model for the DTCs to include private sector participation, and to offer trainees with vouchers for reduced cost training, to be redeemed at DTCs.
- (f) Facilitate knowledge exchange to learn from developed one-stop service centers such as the Kenya's *Huduma* centers and Rwanda's *Irembo* franchise model.

30. **Subcomponent 3.2: Nurturing employment-ready advanced digital skills.** This subcomponent seeks to develop a critical mass of skilled Zambian workers with advanced digital skills and capabilities, enabling them to harness the potential



of emerging technologies. A DTC has been established at ZAMREN within the University of Zambia. This subcomponent will provide support to operationalize the ZAMREN DTC as a hub for advanced digital skills training. Training programs will be tailored to equip university faculty, and government officials, with advanced digital proficiency and students with readiness for the modern job market. To enhance the impact of the technical skills, trainings will be complemented with soft skills to improve collaboration, communication, problem solving, and so on. This subcomponent will include the following activities:

- (a) Conduct a comprehensive digital skills supply and demand gap assessment, with emphasis on advanced digital skills and future needs. This assessment will be based on interviews with industry stakeholders to determine the current availability of digital skills, the specific digital skills in demand by the private sector, and the skills currently being developed in universities and training centers. Additionally, the assessment will focus on the demand for remote workers for e-lancing and from global firms, identifying opportunities for Zambian workers to integrate into global digital value chains, and long-term tracking of those receiving training to track employability.
- (b) Provide training on advanced digital skills based on the gap assessment, at the DTCs, related to emerging technologies such as big data, AI, blockchain, and robotics. This training aims to strengthen the capacity of the Zambian economy to build local solutions and engage in global value chains, to leverage these technologies for economic development. Procedures for selection of trainees will be set out in the Project Implementation Manual (PIM). Training courses will be tailored to the beneficiaries, such as youth, women, elderly, teachers, civil servants, leaders etc.
- (c) Offer customized digital skills and online safety training for women, at the DTCs and via the Digital Ambassadors, along with advanced certified courses aligned with private sector needs. Incorporate rigorous assessment-based certification, mentorship programs featuring female tech role models, and community engagement initiatives to address social norms and foster inclusion.
- (d) Build digital skills and capabilities identified in the National AI Strategy.

Component 4: Program management and capacity building (US\$7 million in IDA grant)

31. **This component will support the critical building blocks for strong implementation, learning, and coordination of the Program.** It will finance the establishment of a Project Implementation Unit (PIU) within SZI and support project management and implementation of project-associated activities. This includes staffing and operating costs of the PIU, independent audits and quality assurance, and monitoring and evaluation (M&E) to support implementation and ensure compliance with Environmental and Social Standards (ESS).

Legal Operational Policies	Triggered?
Projects on International Waterways OP 7.50	No
Projects in Disputed Area OP 7.60	No
Summary of Screening of Environmental and Social Risks and Impacts	

32. **The project will have a nationwide footprint, with efforts targeted at expanding access to affordable and inclusive internet and digitally enabled services.** Through the project, national and regional networks will be expanded, including last mile connectivity for at least 500 selected national sites which will include schools, HEIs and, later, health



facilities, among others. The project has a pro-green approach focused on strengthening regulation for the management of e-waste and applying a dig-once policy to reduce negative impacts on communities and the environment. The project has adopted an inclusive approach that is intended to ensure that project benefits accrue to residents across the country, including rural areas.

33. **An assessment of the project reveals potential environmental and social risks and impacts and occupational health and safety (OHS) risks that may result from proposed activities.** These could include: (a) excavation of trenches for installation of fiber optic cables; (b) minimal vegetation loss; (c) generation of e-Waste; (d) exposure to electrical hazards, including working under high-voltage supply lines; (e) traffic-related incidents; (f) transmission of communicable diseases; (g) potential exclusion of vulnerable groups from accessing project benefits; (h) potential incidences of sexual exploitation and abuse or sexual harassment (SEA/SH) due to contractor-community interactions; (i) potential loss of data privacy due to exposure of personal information to unauthorized individuals; and (j) theft of fuel or oil and solar equipment powering ICT infrastructure if not properly secured. Measures to mitigate these risks have been incorporated in the Environmental and Social Management Framework (ESMF), which was disclosed by SZI on its website on November 29, 2024.³¹ Site-specific Environmental and Social Management Plans (ESMPs), including measures to manage SEA/SH risks, and a Labor Management Procedures (LMP) aligned with the World Bank Environmental, Health Safety Guidelines and Good International Industry Practices, will be developed. The project has prepared and disclosed a Stakeholder Engagement Plan (SEP)³² and an Environmental and Social Commitment Plan (ESCP)³³ with actions and timelines for achieving compliance. TA activities involving capacity building are likely to generate downstream environmental and social risks and impacts. Mitigation measures will include reviewing TORs and AWPBs to ensure activities are conducted in line with the Environmental and Social Framework (ESF) requirements and using alternative technologies, such as satellite, where trenching is not required.

E. Implementation

Institutional and Implementation Arrangements

34. As this is a new project in a country that has not benefitted from any lending in the digital field for many years, a new PIU will need to be established. **Following discussions with the Government, a dedicated PIU for DZAP will be established and operationalized at SZI.** Established in 2015, SZI is well suited to play this central role, given its mandate and track record of spearheading cross-cutting government digitalization initiatives. The functions of the PIU will cover project coordination, procurement, financial management (FM), environmental and social, M&E, communications, and gender. The project beneficiaries will include MoTS and SZI as implementing partners for Subcomponent 1.1; SZI, ZICTA, DPC, the Cybersecurity Agency and ZAMREN for Subcomponents 1.2 and 1.3; MoHAIS and MoTS as implementing partners for Subcomponent 2.1; SZI, MoTS and ZICTA for Subcomponent 2.2; the Ministry of Commerce, Trade and Industry (MCTI), MoTS, MoHAIS, and ZRA as implementing partners for Subcomponent 2.3; and ZAMREN, SZI, ZICTA, ZAMREN and MoTS for Subcomponents 3.1 and 3.2. Further details are provided in the PIM.

35. **The SZI PIU will lead implementation of the project's subcomponents and oversee core project-related fiduciary functions, including FM, procurement, project-related M&E, and environmental and social risk management.** It will also consolidate overall project reporting, working closely with all stakeholders. The PIU will be staffed with necessary experts, including a dedicated Project Coordinator and FM, procurement, environmental and social, and M&E specialists. Additionally, technical specialists in various project areas will liaise with core technical partners on a daily basis.

³¹ https://www.szi.gov.zm/wp-content/uploads/2024/11/DZAP_Environmental-and-Social-Management-Framework_Draft_28.11.2024.pdf

³² https://www.szi.gov.zm/wp-content/uploads/2024/11/DZAP-Stakeholder-Engagement-Plan_Draft.pdf

³³ https://www.szi.gov.zm/wp-content/uploads/2024/11/DZAP-Environmental-and-Social-Commitment-Plan_28.11.2024.pdf



Recruitment for most of the envisioned PIU staff is ongoing, with support from the PPA. The project will ensure that the hired technical experts transfer their knowledge and provide hands-on training to SZI staff to ensure sustainability after project completion.

36. **SZI will closely collaborate with several MDAs as key stakeholders, technical partners, and main project beneficiaries for planned investments, and these MDAs have been actively involved in shaping the project design.** The project beneficiaries will include the MoTS, MoHAIS, SZI, DPC, Cybersecurity Agency, ZAMREN, MCTI, ZICTA, Ministry of Education and, later, Ministry of Health for connecting hospitals and health centers. These MDAs are expected to contribute technical inputs and evaluation of bids while working closely with the technical specialists embedded in the PIU. They will support SZI on technical aspects of procurement-related processes, support quality assurance and validation of key deliverables, and provide input needed for effective M&E framework management and reporting. Detailed roles and responsibilities and terms of engagement will be captured in the PIM, to be finalized before project effectiveness. Memoranda of Understanding (MoUs) will be established between the PIU and partner MDAs at the start of the project implementation, as needed.

37. **A dedicated Project Steering Committee (PSC), to be chaired by the Secretary to the Cabinet, will be established to provide strategic oversight and governance for the project,** with the National Coordinator of SZI serving as its secretary. The PSC's mandate will include: (a) approval of Annual Work Plan and Budget (AWPB); (b) quarterly review of project progress; and (c) provision of strategic guidance and recommendations related to project implementation. The PSC will include representatives from key government institutions and project beneficiaries.

38. **The PSC will be supported by one or more targeted technical committees (TCs), as needed, to address any key *ad hoc* technical issues that arise and provide operational support.** These dedicated component-specific TCs will convene relevant government MDAs deemed essential for successful implementation and stakeholder management and support key technical-level decisions requiring broader agreement. TCs may include representatives from the private sector, academia, and civil society. TC meetings will be chaired and called by relevant SZI PIU technical specialists. TCs can escalate issues to the PSC if they require strategic discussion. The terms of reference for TCs, including expected membership, is detailed further in the PIM.

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